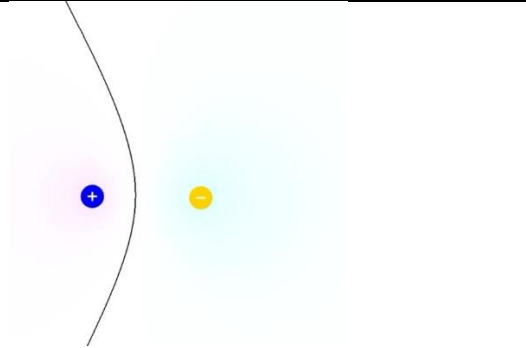


| Qn |     | Answer                                                                                                                                                                                                             | Mark              |
|----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| 5  | (a) | <ul style="list-style-type: none"> <li>electric field due to point charges are radial.</li> </ul>                                                                                                                  | B1                |
|    |     | <ul style="list-style-type: none"> <li>So vectors can only cancel away completely if they are collinear / parallel. Hence ...</li> </ul>                                                                           |                   |
|    |     | Alternative <ul style="list-style-type: none"> <li>electric field due to point charges are radial.</li> </ul>                                                                                                      |                   |
|    |     | <ul style="list-style-type: none"> <li>So vertical components of vectors can only cancels away completely along the line joining the two charges.</li> </ul>                                                       | B1                |
|    | (b) | Show understanding that electric field for point charges $\propto \frac{1}{r^2}$                                                                                                                                   | B1                |
|    |     | Deduce that P is at left of A                                                                                                                                                                                      | B1                |
|    |     | $\frac{1}{4\pi\epsilon_0} \frac{2.4}{x^2} = \frac{1}{4\pi\epsilon_0} \frac{2.9}{(0.15 + x)^2}$                                                                                                                     | M1                |
|    |     | <ul style="list-style-type: none"> <li><math>x = 1.51 \text{ m}</math></li> </ul>                                                                                                                                  | A1                |
|    |     |                                                                                                                                                                                                                    |                   |
|    |     | Example of alternatives<br><br>define P distance $x$ to be right of B<br>$\frac{1}{4\pi\epsilon_0} \frac{2.4}{(0.15 + x)^2} = \frac{1}{4\pi\epsilon_0} \frac{2.9}{x^2}$ $x = -1.66 \text{ m}$ <u>and</u> left of B | C1<br>B1,M1<br>A1 |
|    | (c) |  <p>B1: line cuts AB nearer to lesser charge<br/>           B1: line curves towards lesser charge.</p>                          | B1 ; B1           |